MATTEO STOCKINGER

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PROFILE

PhD Student at the Max Planck Institute for Gravitational Physics, Potsdam, Germany

LANGUAGES

French (native language)
German C1
English C2
Chinese HSK3
Italian : basic knowledge
Spanish : basic knowledge

AREAS OF INTEREST

Reading
Music (piano, organ, cello, viola da gamba)
Languages
Linguistics
Traveling
History
Volley-ball
Football

SUMMARY/RESEARCH PROFILE

PhD Candidate in the Computational Relativistic Astrophysics Group at the Max Planck Institute for Gravitational Physics. Specialized in developing novel mathematical and machine learning (PINNs) methods for relativistic astrophysics.
Proven track record of high-level interdisciplinarity combining theoretical physics, fundamental physics, and high-performance computing (HPC)

EDUCATION

-2023-2027 (Expected) Max Planck Institute for Gravitational Physics (Albert Einstein Institute)
PhD Candidate in Physics,
Dissertation: « New methods in GR » under the supervision of Prof. Masaru Shibata
Focus: Relativistic astrophysics

-2019-2023 École Normale Supérieure (ENS Ulm), Paris, France
Diploma of the École Normale Supérieure (DENS)
France's most prestigious institutional degree program (top 0.1% nationwide selectivity).
Interdisciplinary Curriculum: Pursued advanced graduate-level courses in Physics and Mathematics, alongside parallel training in Biology, Cognitive Science, Economics,
Comparative Linguistics, and Chinese (HSK3),

-2022-2023 Sorbonne Université
Master 2 (M.S.) of fundamental mathematics

Master thesis under the supervision of Prof. Mihai Tibăr: « Singularity theory », Université de Lille

-2020-2022- Ecole Normale Supérieure ULM Paris
Master ICFP (M.S.) (International Center of Fundamental Physics), University of Paris-Sud

Master 2 Internship on granular materials at the LPS, Université de Paris-Saclay, under the supervision of Prof. Giuseppe Foffi: « Study of a phase transition in granular materials »
Master 1 internship in Numerical Relativity at the Max Planck Institute for Gravitational Physics (Albert Einstein Institute) in Potsdam, under the supervision of Prof. Masaru Shibata : « Neutron Stars in Massive Scalar Tensor Theory »

-2019-2020 Ecole Normale Supérieure ULM Paris
Double Bachelor Degree (B.S.) in Mathematics and Physics, University of Orsay-
Paris-Sud
Completed the full standalone rigorous curricula of both majors simultaneously.

Bachelor Thesis about Random Matrix Theory
Bibliographical research at the ENS Ulm Paris under the supervision of Dr. G. Barranquand
Internship at the Erbium Lab, University of Innsbruck, Austria under the supervision of Dr. L. Chomaz

-2019 Highly Selective National Graduate School Entrance Examinations (Concours)
Ranked among the top nationwide applicants; successfully admitted to France's most elite institutions: **ENS Ulm**, **École Polytechnique** (l'X), **ESPCI**, **ENS Cachan** (Paris- Saclay), and **ENS Lyon**

-2017-2019 Classes Préparatoires aux Grandes Écoles (CPGE) — PC*/PCSI
Ranked among the top 3 elite French institutions; highly intensive STEM curriculum preparing for national graduate school entrance examinations (Focus: Advanced Mathematics and Physics).
Two-year intensive, elite STEM undergraduate curriculum focusing on advanced Mathematics, Physics, and Chemistry.

PUBLICATIONS

Resonances of compressible stars in precessing orbits around a spinning black hole,
M. Stockinger, M. Shibata, Physical Review D 111, 123044 (2025)

Tame deformations of highly singular function germs,
C. Joiţa, **M. Stockinger**, M. Tibăr, Journal of Geometry and Physics, 213, 105471 (2025)

Relativistic Roche problem for stars in precessing orbits around a spinning black hole,
M. Stockinger, M. Shibata, Physical Review D 110, 043038 (2024)

Interplay between an Absorbing Phase Transition and Synchronization in a Driven Granular System,
R. Maire; A. Plati; **M. Stockinger**; E. Trizac; F. Smallenburg; G. Foffi, Physical Review Letters, 132, 238202 (2024)

PRESENTATIONS

Nov. 2026 Invited talk at EPaCo group, University of Dartmouth, USA
« PINNs methods for the study of the magnetosphere of compact objects »

Sep. 2026 contributed talk CoCoNut Meeting, Southampton, England
« PINNs methods for the study of the magnetosphere of compact objects »

Feb. 2026 poster Multi-Messenger Astrophysics in the Dynamic Universe, Kyoto, Japan
« PINNs methods for the study of the magnetosphere of compact objects: Preliminary results »

July 2025 contributed talk, GR/Amaldi 25, Glasgow, Scotland
« Resonances of compressible stars in precessing orbits around a spinning black hole »

April 2025 contributed talk, DPG-Spring meeting 2025, Göttingen, Germany
« Resonances of compressible stars in precessing orbits around a spinning black hole »

TEACHING EXPERIENCE

2024, Gravitational Waves Astrophysics,
Teaching Assistant, University of Potsdam

OUTREACH

May 2025: Potsdam Science Day

DISTINCTIONS

-Gold Medal for France at the 2018 International Chemistry Olympiad (IChO) in Prague/Bratislava
-Laureate 2018 of the Académie des Sciences IChO

COMPUTER SKILLS

Programming in Python (Jax, PyTorch), in C
OS : Windows, Linux + HPC experience